Infections, Diseases and Immunity

BIO101
LECTURE 06

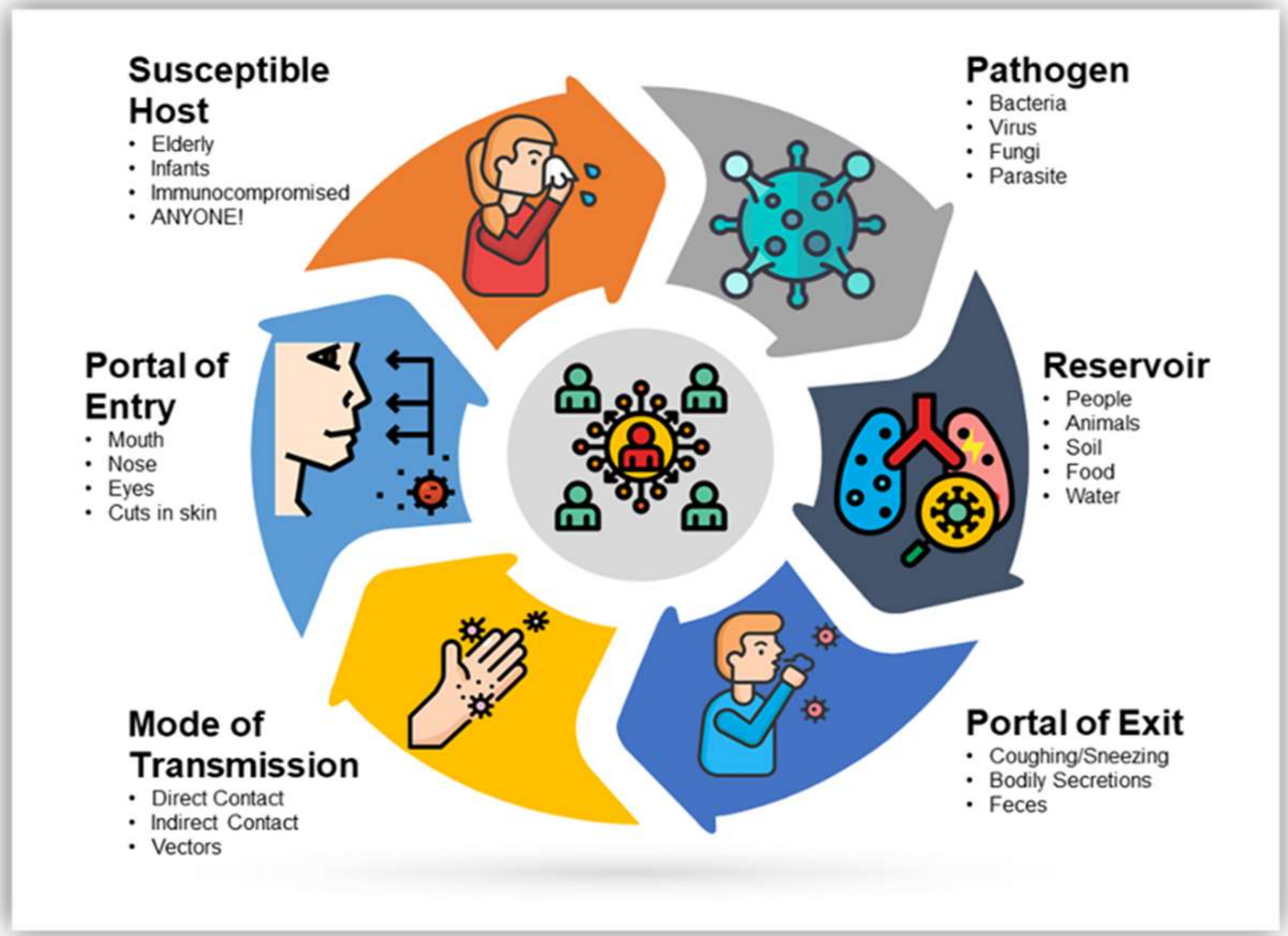
What is infection?

- **Infection** is the invasion and multiplication of harmful microorganisms (such as bacteria, viruses, fungi, or parasites) in the body.
- These microorganisms are called **pathogens**.
- Infection may lead to tissue damage and cause illness.
- It can be **localized** (in one area) or **systemic** (spread throughout the body).

Bacterial infection

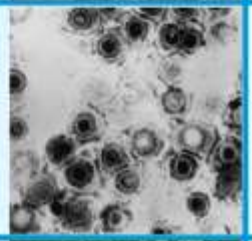
Fungal infection

Chain of infection



Pathogens

- A pathogen is a living thing that causes disease.
- Pathogenicity refers to the ability of a pathogenic agent to cause disease.

Type of pathogen	Description	Human diseases caused by pathogens of that type
Bacteria *Escherichia coli*	Single-celled organisms without a nucleus	Strep throat, staph infections, tuberculosis, food poisoning, tetanus, pneumona, syphilis
Viruses *Herpes simplex* 	Non living particles that reproduce by taking over living cells	Common cold, flu, genital herpes, col sores, measle, AIDS, genital warts, chicken pox, small pox
Fungi *Death cap mushroom*	Simple organisms, including mushrooms and yeasts, that grow as single cells or thread like filaments.	Ringworm, athlete's foot, tineas, candidiasis, histoplasmosis, mushroom poisoning
Protozoa *Giardia lamblia*	Single-celled organism with a nucleus.	Malaria, "traveller's diarrhea" giardiasis, typano somiasis ("sleeping sickness")

Reservoir

It refers to any human being, animal, plant, soil, or inanimate matter in which a parasite lives, multiplies, and depends for its survival.

Sources of infection:

1. Endogenous sources:

Organisms of normal flora that are usually non-pathogenic but occasionally behave as pathogens outside their habitat.

2. Exogenous sources:

a. ***Human***

b. ***Animals***

c. ***Insects***

d. ***Soil and Water***

e. ***Foods***

Reservoir

Exogenous sources:

a. Human cases and carriers: The Commonest source of infection is human.

i. **Healthy carrier:** harbors the pathogen but never suffered from the disease caused by a particular pathogen.

ii. **Convalescent carrier:** one who has recovered from the disease but continues to harbor the pathogen in his body.

iii. **Temporary carrier:** this carrier state lasts for about 6 months.

iv. **Chronic carrier:** harbors the pathogen for several years and sometimes for the rest of one's life.

v. **Paradoxical carrier:** A person who acquires the pathogen from another carrier.

vi. **Contact carrier:** one who acquires the pathogen from a patient.

Reservoir

<u>Exogenous sources:</u>

b. Animals

Infectious diseases transmitted from animals to man are called zoonoses.

- Zoonotic diseases may be:
- Bacterial – bovine tuberculosis
- Viral- rabies from dogs
- Protozoal – Toxoplasmosis from cats
- Helminthic – Taeniasis from cattle
- Fungal- dermatophytes from dogs

Reservoir

Exogenous sources:

c. **Insects**

Diseases transmitted by insects are called arthropod-borne diseases.

Insects transmitting diseases are called vectors.

- **Mechanical vectors:** carry the organisms on their wings, legs, and body. Eg transmission of typhoid bacilli to man through food by domestic flies.
- **Biological vectors:** the pathogen multiplies in the body of the vector. E.g. female anopheles mosquito in malarial parasite.
- **Extrinsic incubation period:** after the entry of the pathogen in the vector the time required for the vector to become infective.

d. Soil and Water

- Some pathogens survive in the soil for long periods. E.g. spores of tetanus and gas gangrene bacilli, the fungus causing histoplasmosis, and parasites such as roundworm and hookworm.
- Contamination of water with vibrio cholera and hepatitis virus act as the source of infection.

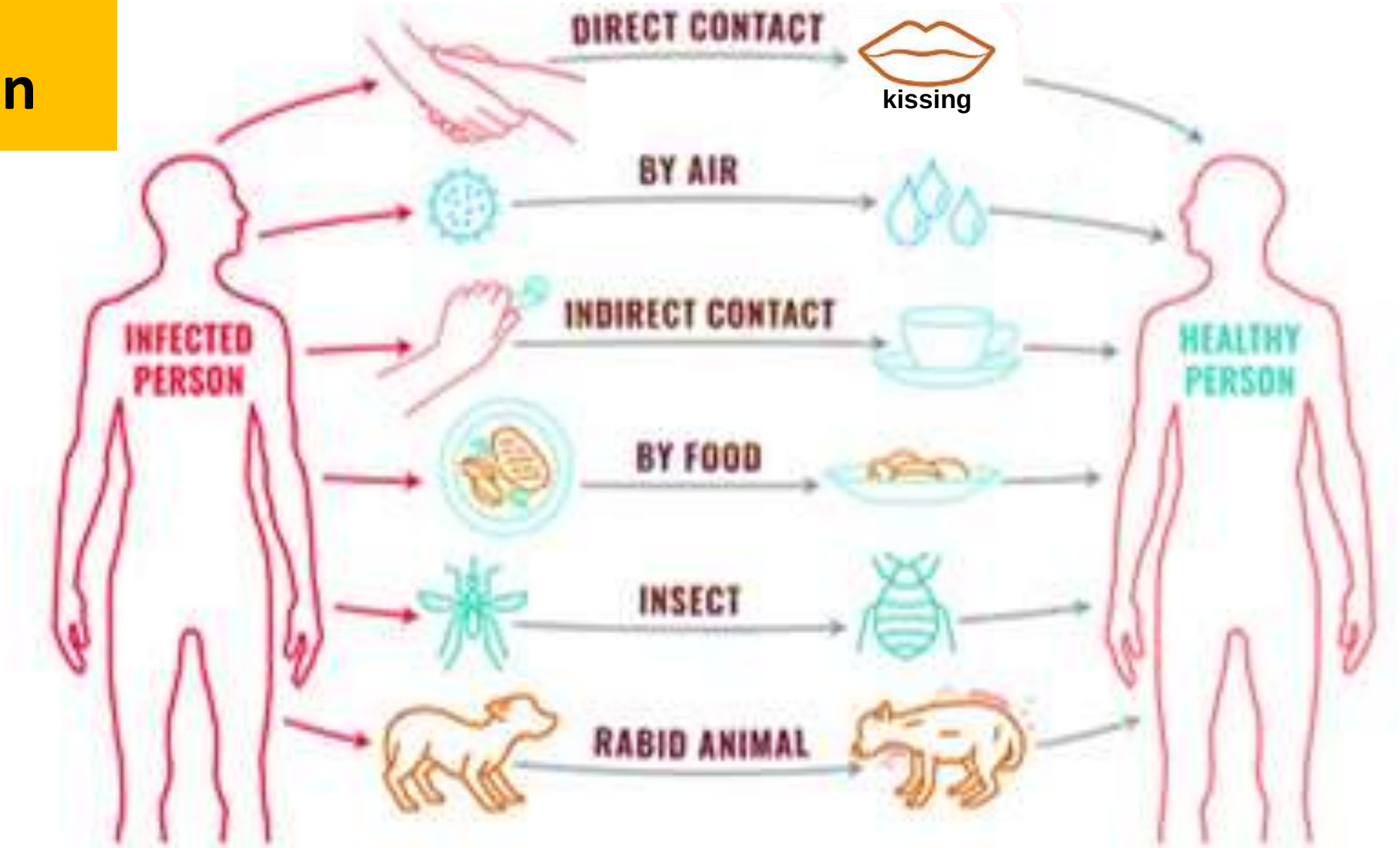
Reservoir

<u>Exogenous sources:</u>

e. Food

Contaminated food acts as a source of infection in the case of food poisoning, gastroenteritis, diarrhea, and dysentery.

Modes Of Transmission



Modes Of Transmission

i. **Contact**

a. Direct contact – directly through physical contact. STDs such as syphilis, gonorrhea, herpes simplex type 2, and AIDS.

b. Indirect contact- indirectly acquired through fomites.

ii. **Airborne transmission:**

a. Droplets of respiratory infection are spread by inhalation.

b. Droplet nuclei (1-10 micron diameter) remain airborne as aerosols act as a source of infection.

iii**. Ingestion**

Intestinal infections like cholera, dysentery, food poisoning, etc are acquired by ingestion of food or drink contaminated by pathogens.

Occurs mostly through carriers engaged in food handling or contaminated water supply.

Modes Of Transmission

iv. **Inoculation**

Direct inoculation in the tissues e.g. rabies virus inoculation through dog bite or spores of *Clostridium* leading to tetanus.

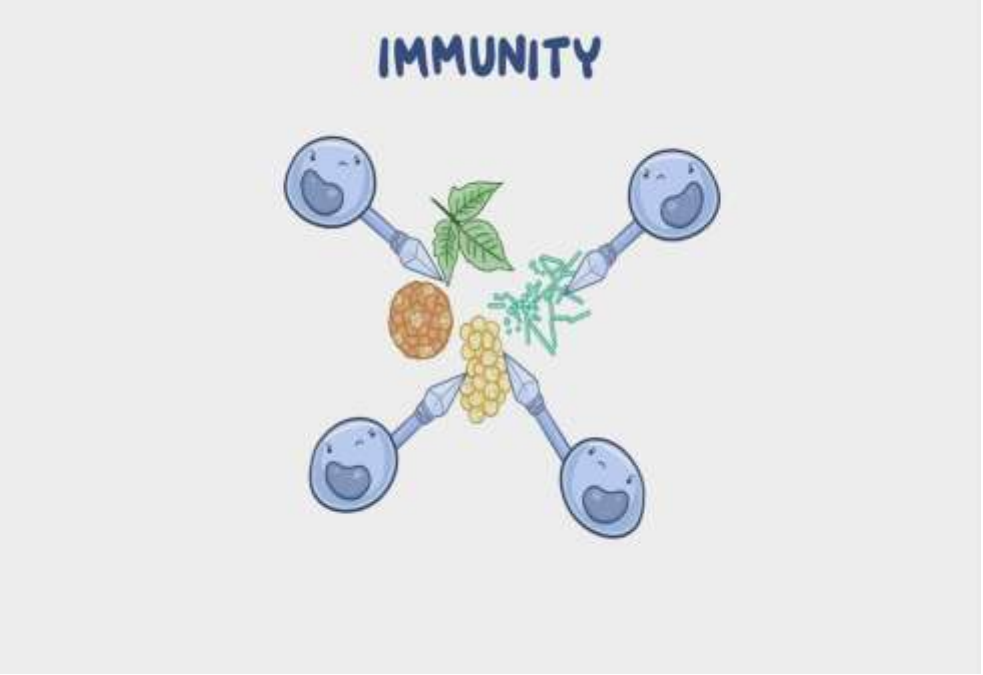
v. **Transplacental**

Transmission of the pathogen from mother to fetus via placenta. Eg rubella virus, toxoplasma, CMV. Also known as vertical transmission.

vi**. Iatrogenic infection**

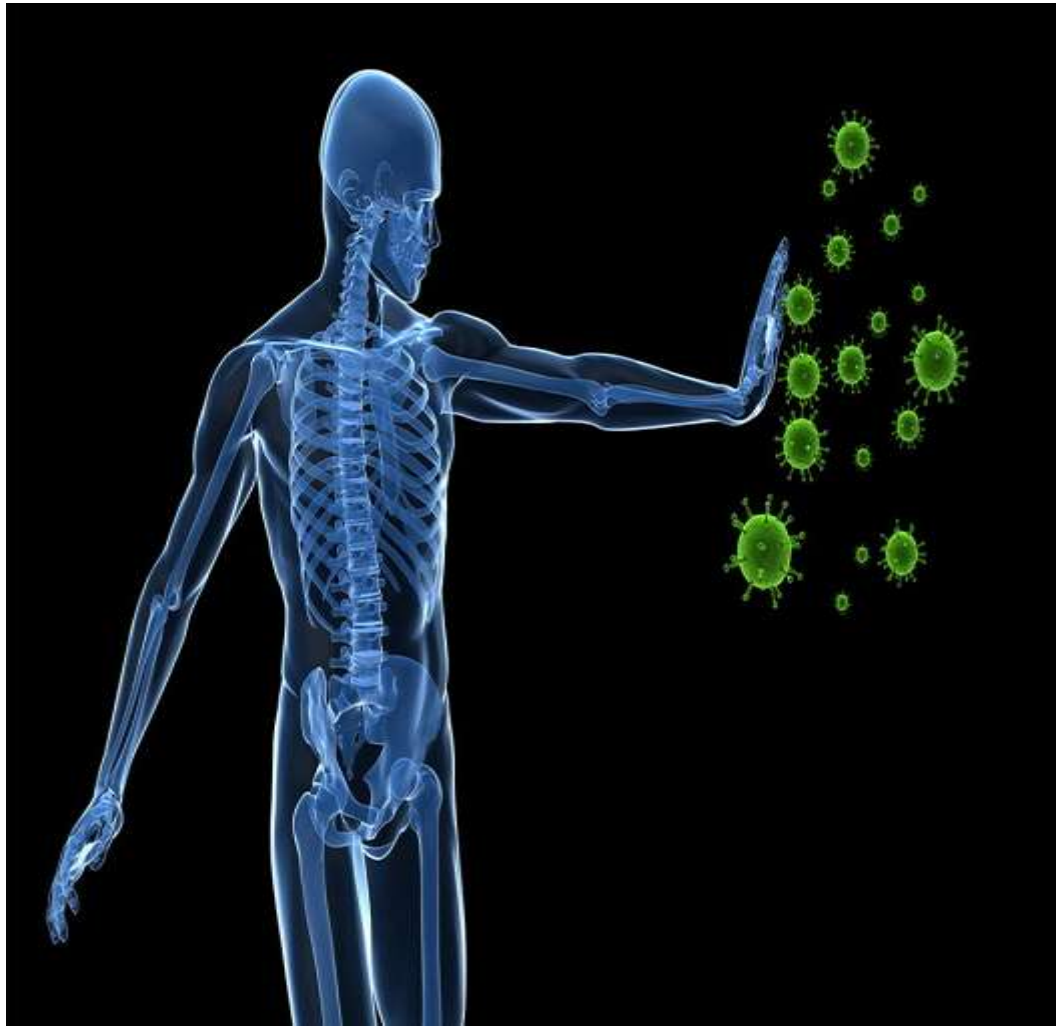
Infections like AIDS and Hepatitis B are transmitted during lab and surgical procedures such as lumbar puncture, blood transfusion, dialysis, and surgery.

Can we fight against disease-causing organisms ?



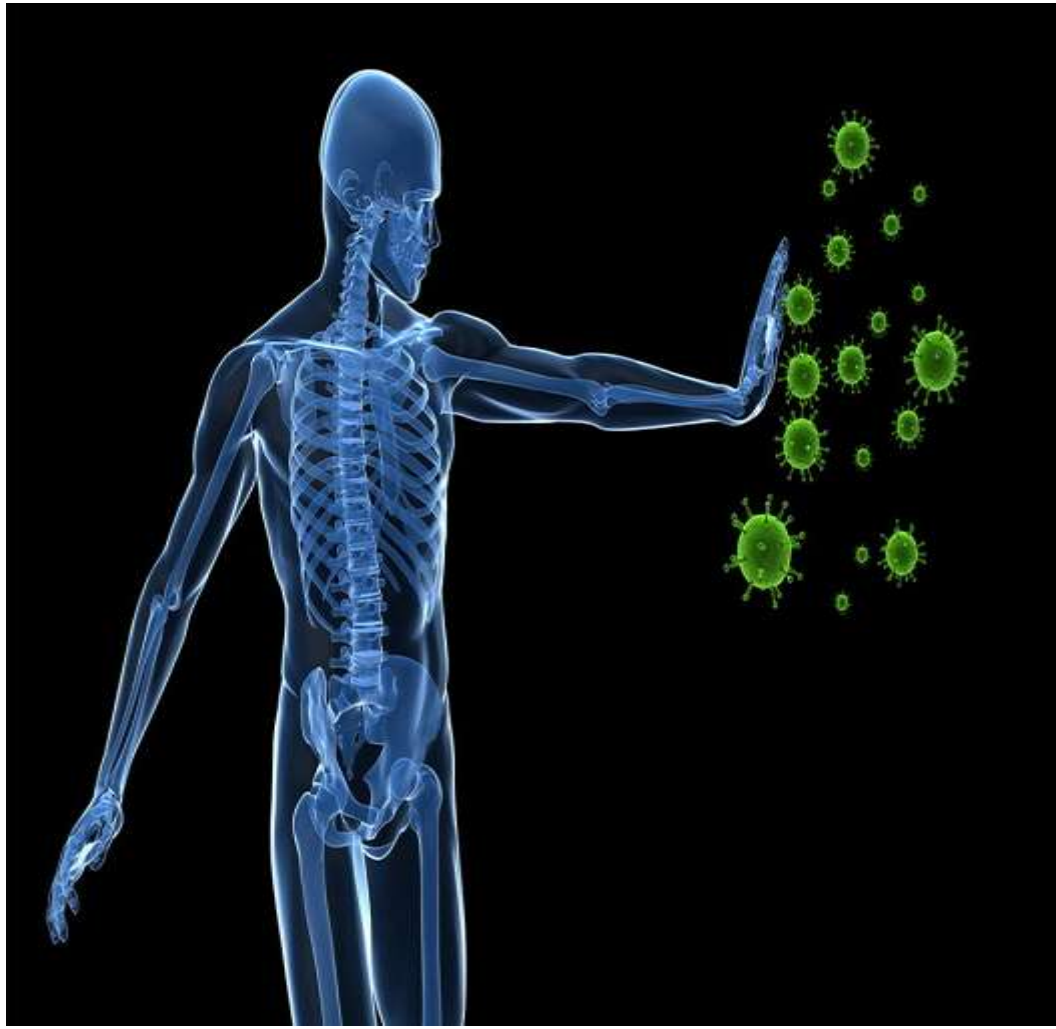
Why Do We Still Need Medicine & Vaccines If We Have an Immune System?

- Pathogen: Load, resistance, variants.
- Host: Age, physical condition, nutrients.
- Outbreaks
- Speed and Timing



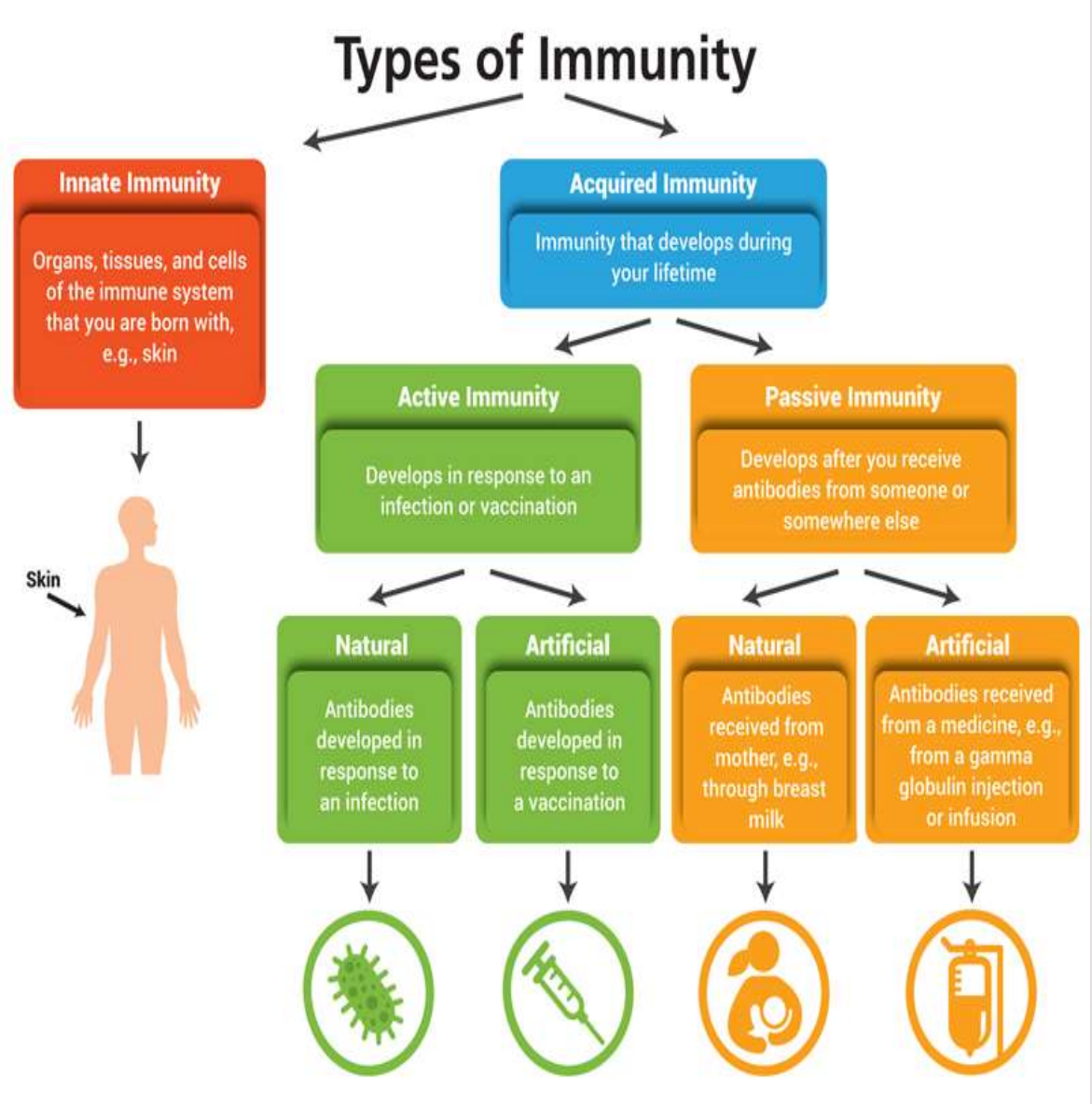
Can we fight against disease-causing organisms ?

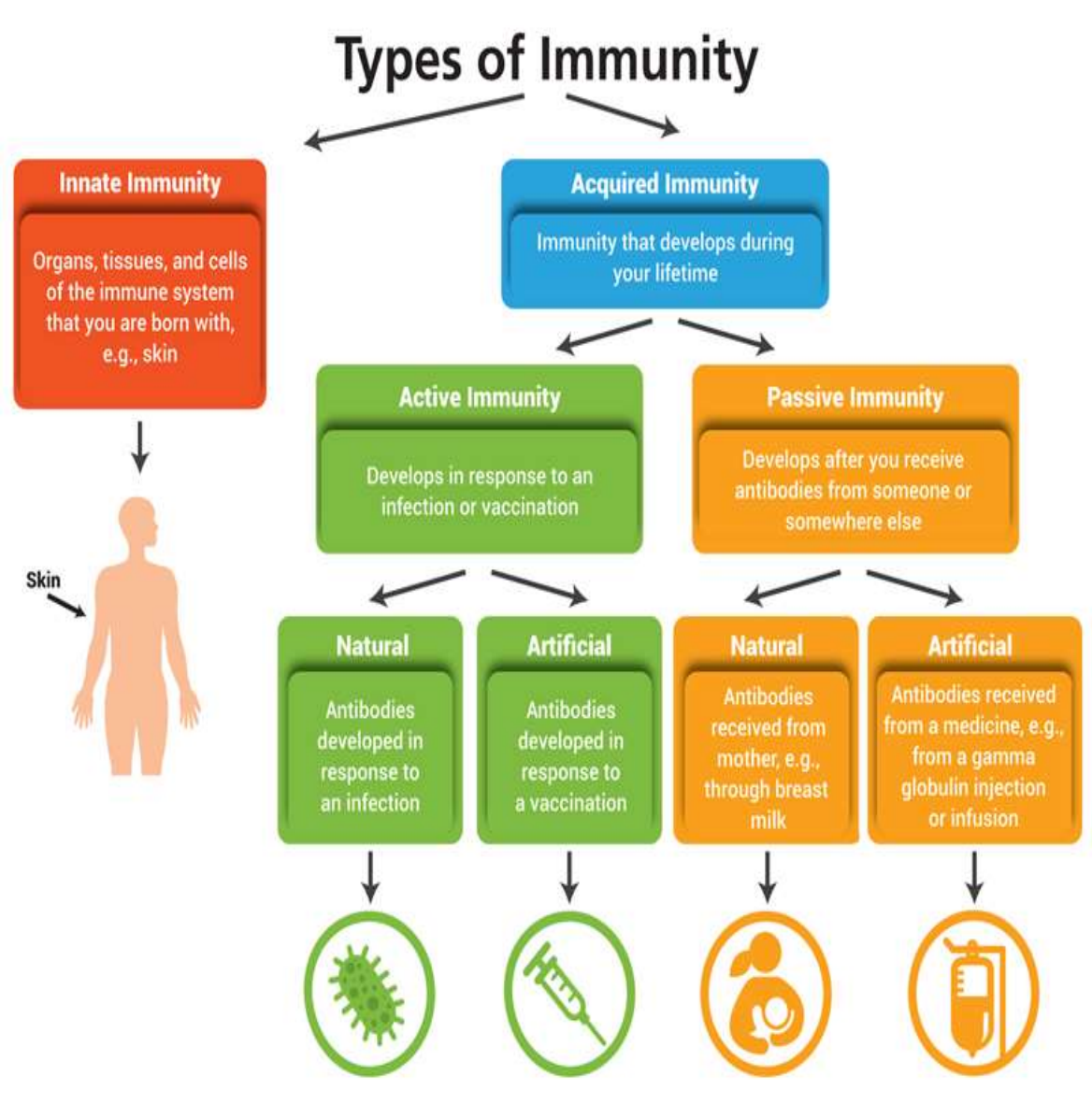
- **Immunity**-The ability of an organism to recognize and defend itself against specific pathogens or antigens.
- The cells and molecules responsible for immunity constitute the **immune system** and their collective and coordinated responses to foreign substances is called **immune response**.

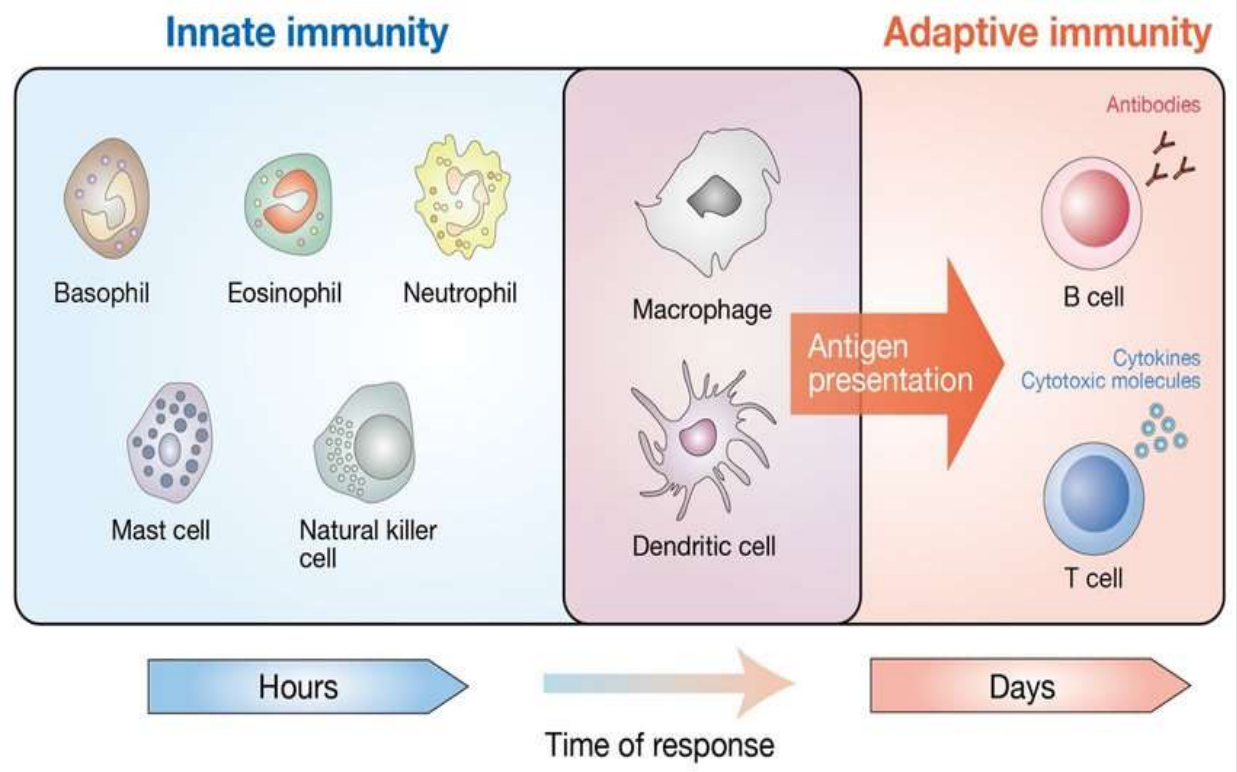


IMMUNITY

- Innate immunity represents an ancient defense system that has evolved to recognize conserved patterns characteristic of a variety of pathogens and often serves as the first line of defense.
- Adaptive immunity, recognizes novel molecules produced by pathogens by virtue of a large repertoire of specific antigen receptors.




Types of Immunity
Innate Immunity
Organs, tissues, and cells of the immune system that you are born with, e.g., skin
Skin
Acquired Immunity
Immunity that develops during your lifetime
Active Immunity
Develops in response to an infection or vaccination
Passive Immunity
Develops after you receive antibodies from someone or somewhere else
Natural
Antibodies developed in response to an infection
Artificial
Antibodies developed in response to a vaccination
Natural
Antibodies received from mother, e.g., through breast milk
Artificial
Antibodies received from a medicine, e.g., from a gamma globulin injection or infusion

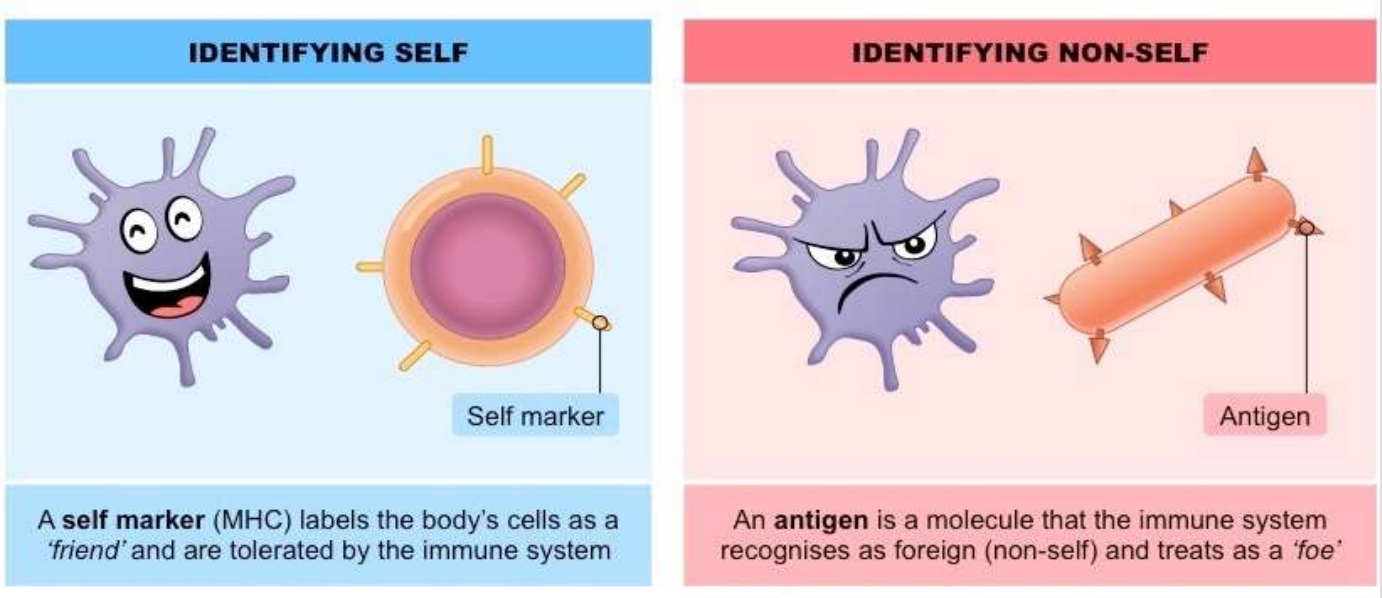

Innate immunity
Adaptive immunity
Basophil
Eosinophil
Neutrophil
Mast cell
Natural killer cell
Macrophage
Dendritic cell
Antigen presentation
Antibodies
B cell
Cytokines
Cytotoxic molecules
T cell
Hours
Days
Time of response

What is self and non-self in the immune system?

The immune system's ability to distinguish "self" (body's own cells) from "non-self" (foreign materials) is crucial for maintaining health. This discrimination prevents the immune system from attacking healthy tissues and allows it to effectively target pathogens.

- **Self-antigens:** Molecules found on the surface of the body's own cells, proteins, and tissues. MHC is a self marker.
- **Non-self antigens:**

Molecules recognized as foreign, such as those on pathogens (bacteria, viruses, fungi) or allergens.



Immune System

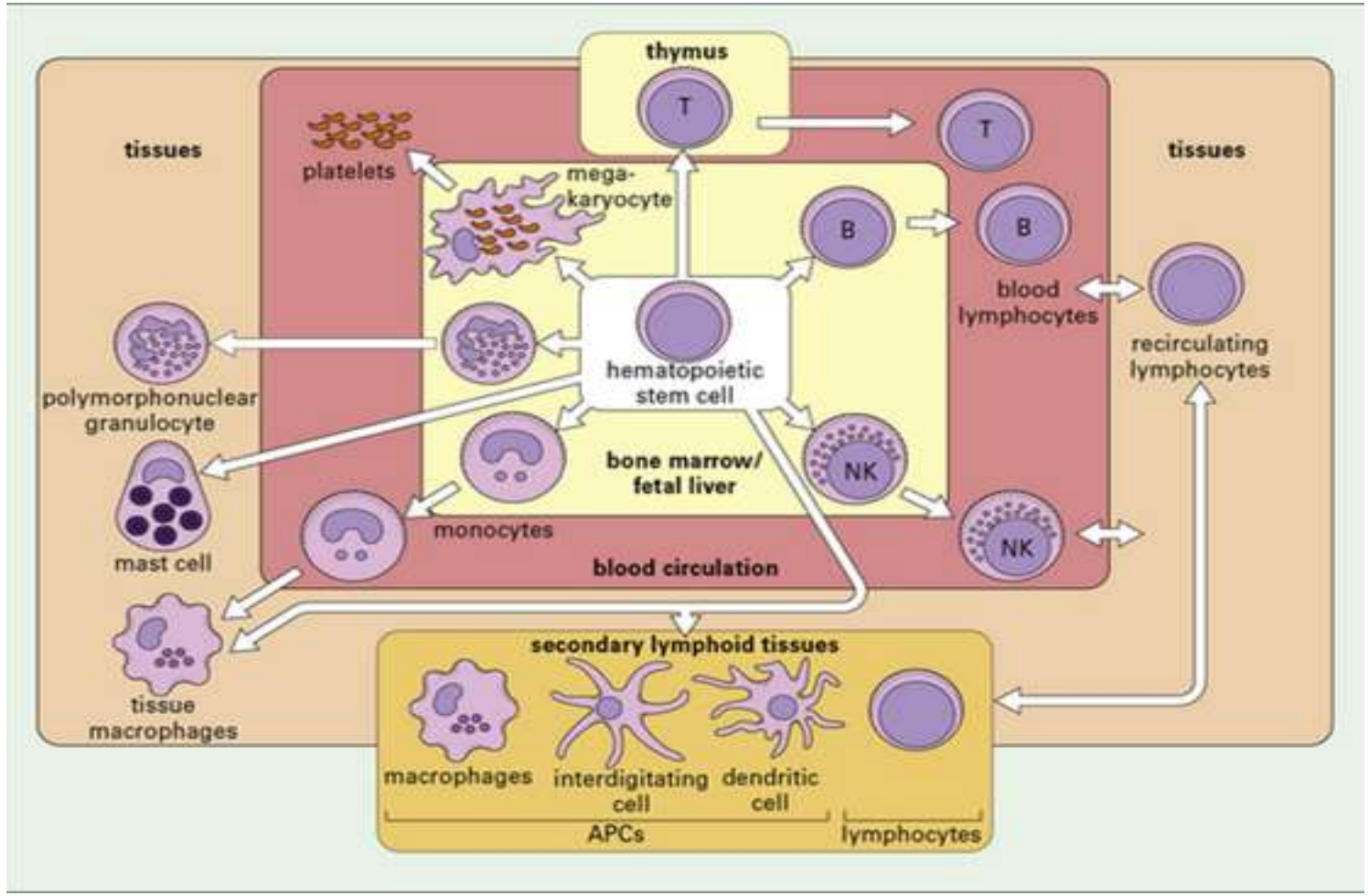
When the immune system is working properly, it:

Tells the difference between cells that are yours and those that don’t belong in your body.

How does immune system work particularly?

- Immune system is a large network of organs, white blood cells, proteins and chemicals. These parts all work together to protect you from germs and other invaders. Our immune system also helps our body heal from infections and injuries. Immune system works hard to keep you healthy. It does this by:
  - Keeping invaders (like germs) out of our body.
  - Destroying invaders.
  - Limiting how much harm the invaders can do if they're inside our body.
  - Healing damage to your body.
  - Adapting the nature of the germs and keep memory of the infection.

Origins of cells of immune system



All cells shown here arise from the **hematopoietic stem cell** in bone marrow.

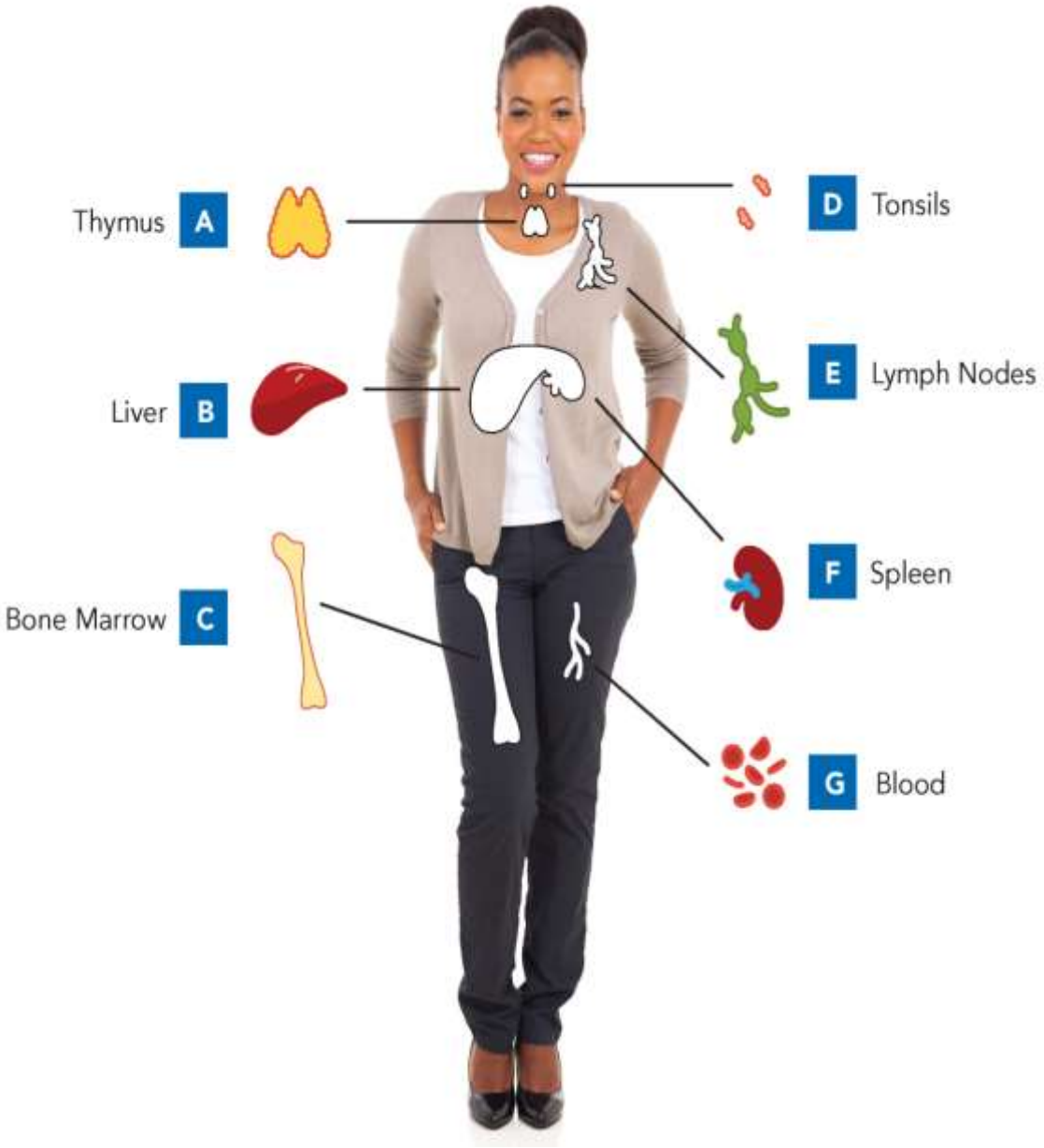
Immune system

White blood cells. These immune system cells attack and eliminate harmful germs to keep you healthy. There are many types of white blood cells, and each type has a specific mission in your body's defense system. Each type also has a different way of recognizing a problem, communicating with other cells and getting their job done.

- **Antibodies**. These proteins protect you from invaders by binding to them and initiating their destruction.
- **Cytokines**. These proteins serve as chemical messengers that tell your immune cells where to go and what to do. Different types of cytokines do different specific tasks, like regulating inflammation. Inflammation happens when your immune cells are warding off invaders or healing damage to your tissues.

What other parts of your body make up the immune system?

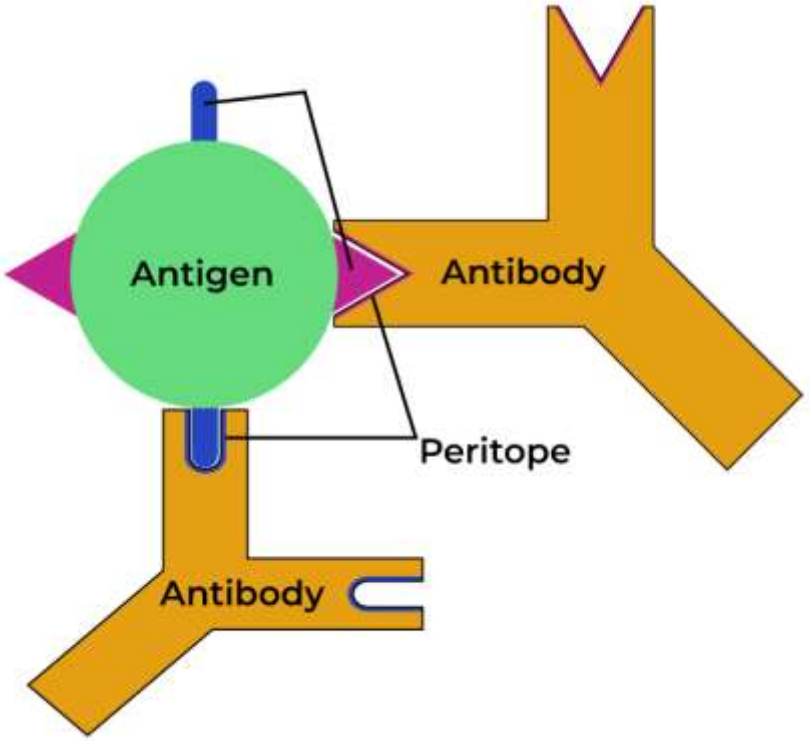
- Lymph nodes
- Spleen
- Tonsils and adenoids
- Thymus.
- Bone marrow
- Skin
- Mucosa
- Complement system



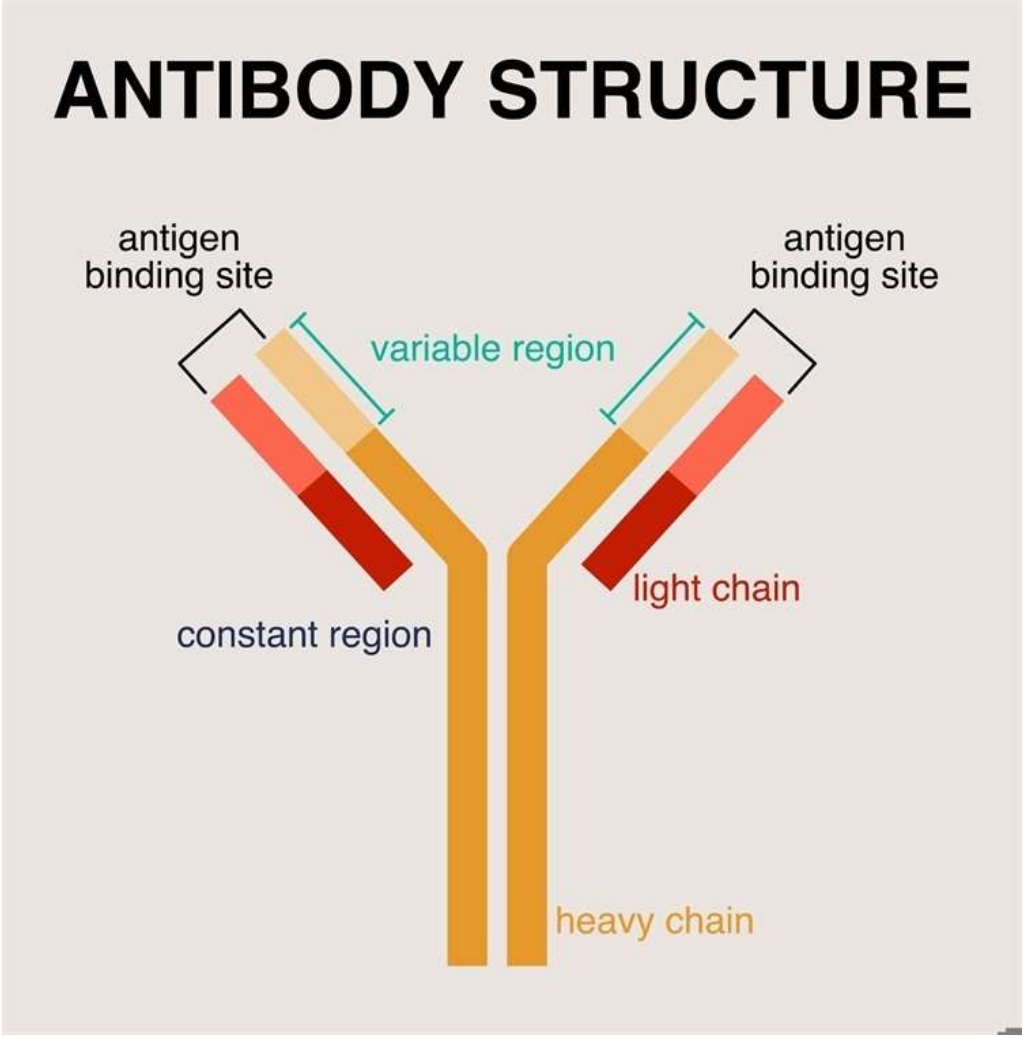
How do antibodies work?

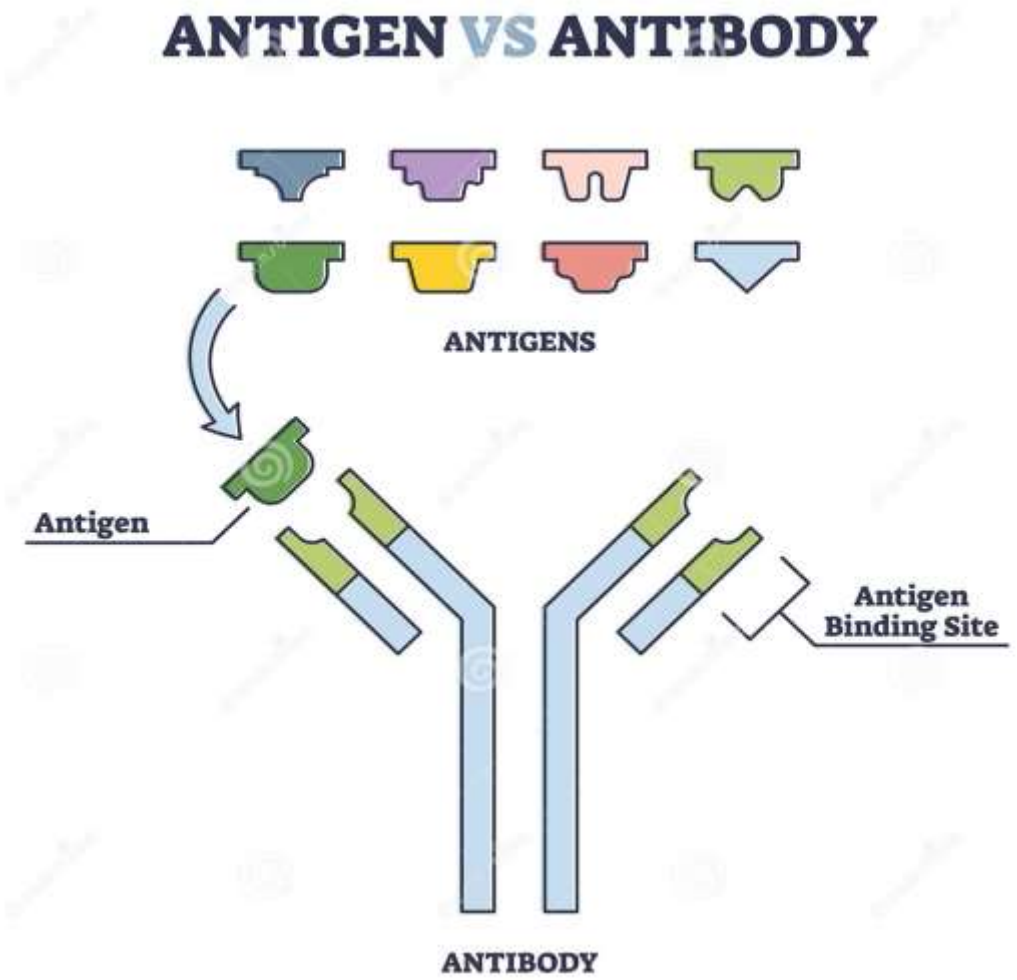
- **An antigen** is unwanted foreign substance that enters our body. Foreign substances can include bacteria, viruses, fungi, allergens, venom, and other various toxins and disease-causing organisms.

- **An antibody** is a protein produced by our immune system to attack and fight off these antigens. Immunoglobulin is used as another word for antibody.

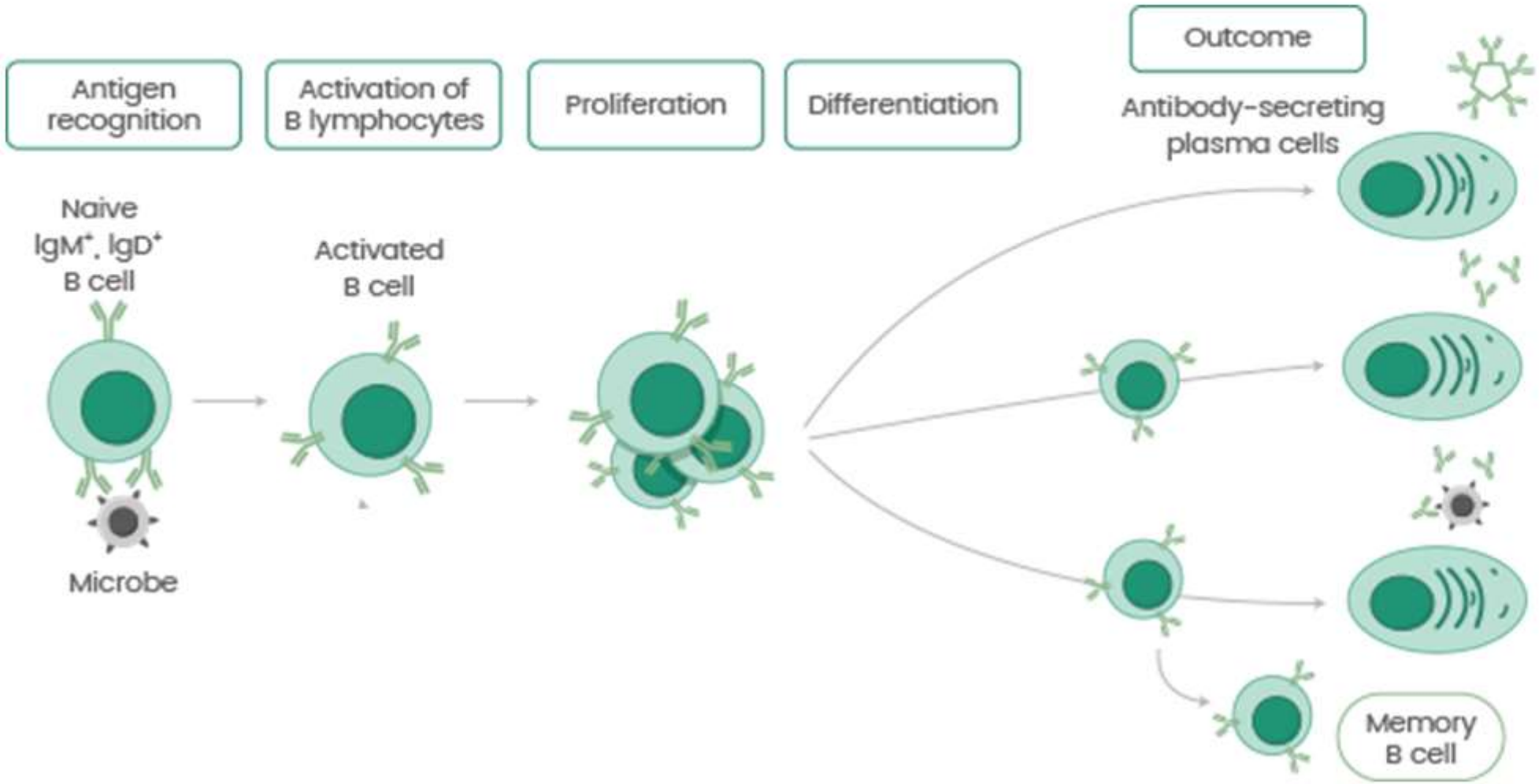


How do antibodies work?





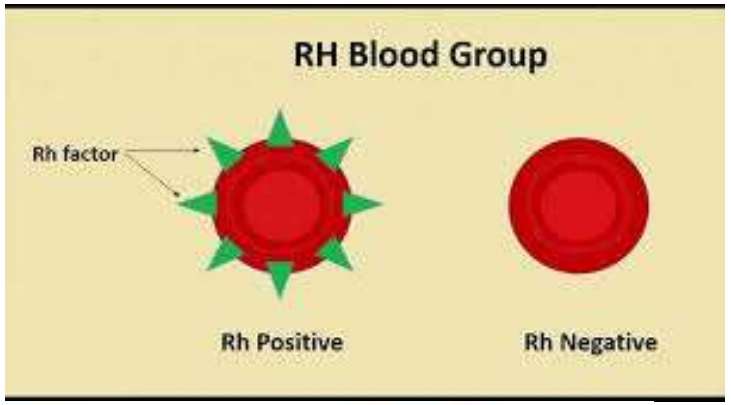
How do antibodies work?

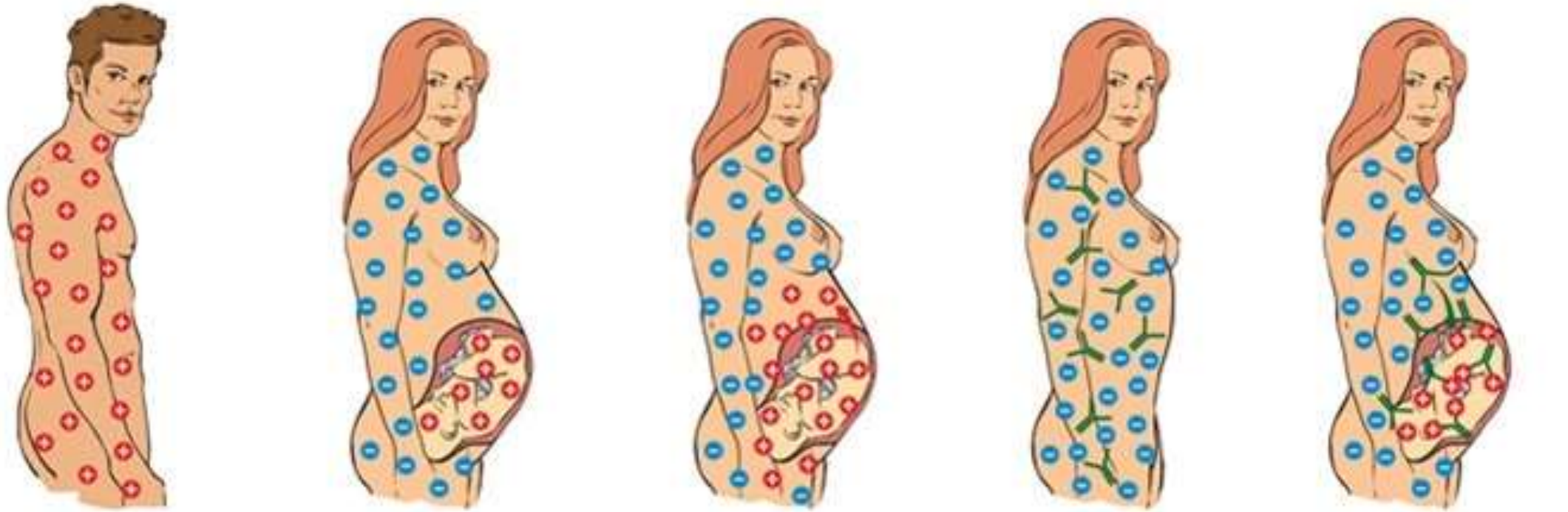


Antibody-Antigen Aspect of Blood Grouping

	Group A	Group B	Group AB	Group O
Red blood cell type	A	B	AB	O
Antibodies in Plasma	Anti-B	Anti-A	None	Anti-A and Anti-B
Antigens in Red Blood Cell	A antigen	B antigen	A and B antigens	None

Blood Grouping





Rh-positive man

Rh-negative woman with Rh-positive fetus

Rh-positive fetus antigens can enter the mother's blood during delivery

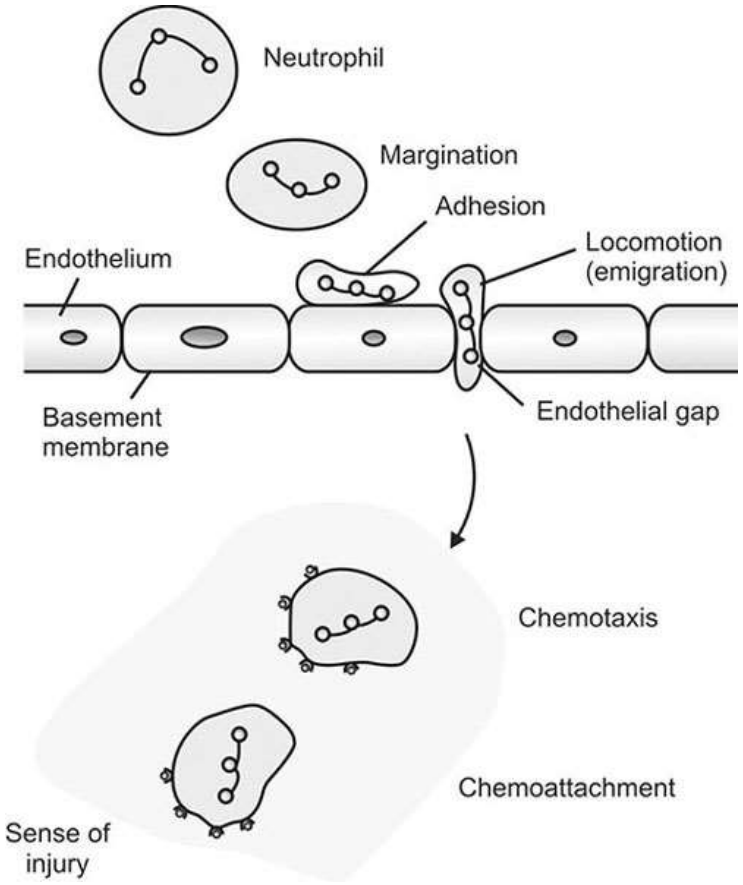
Mother will produce anti-Rh antibodies

In the next Rh-positive pregnancy, mother's anti-Rh antibodies will attack fetal red blood cells

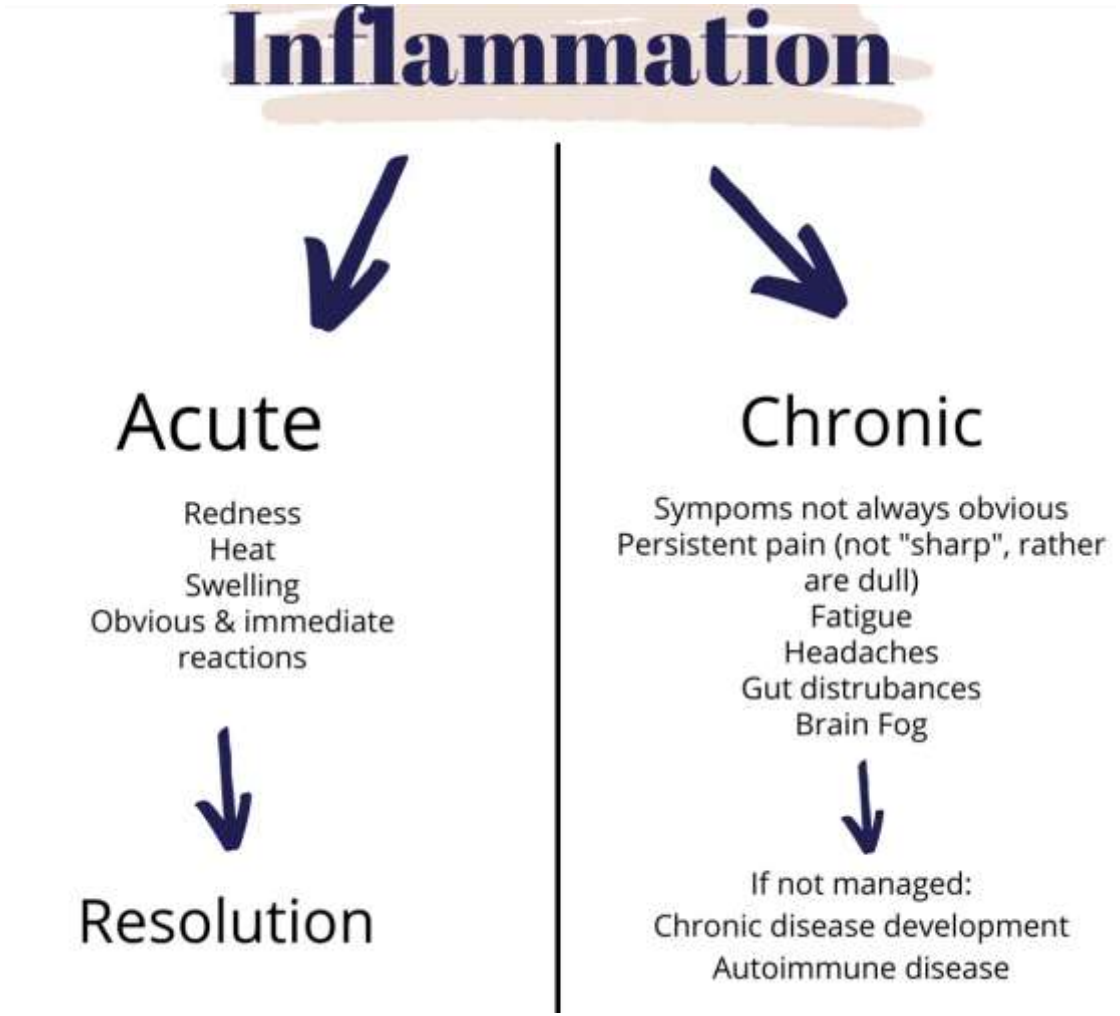
Inflammation

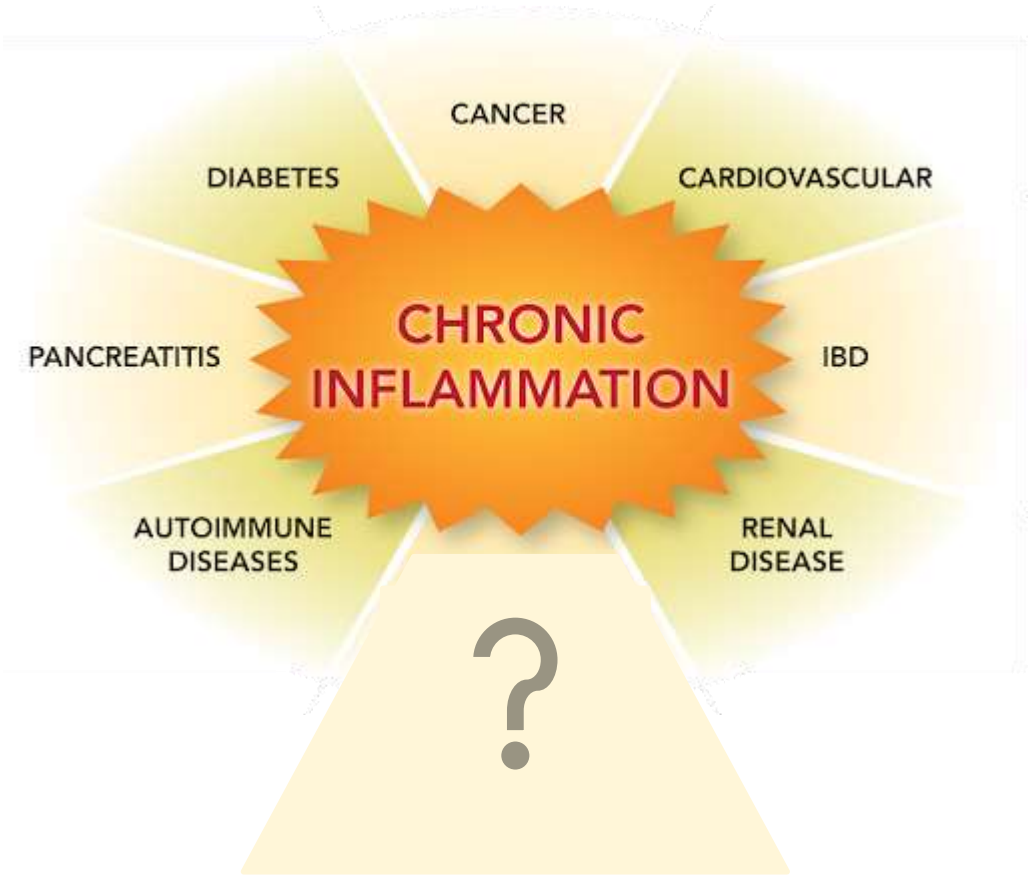
The cells of the immune system are widely distributed throughout the body. But if an infection occurs it is necessary to concentrate them and their products to the site of infection. This process is called inflammation.

- Changes in blood flow,
- An increase in permeability of blood vessels,
- The migration of fluid, proteins, and white blood cells (leukocytes) from the circulation to the site of tissue damage.



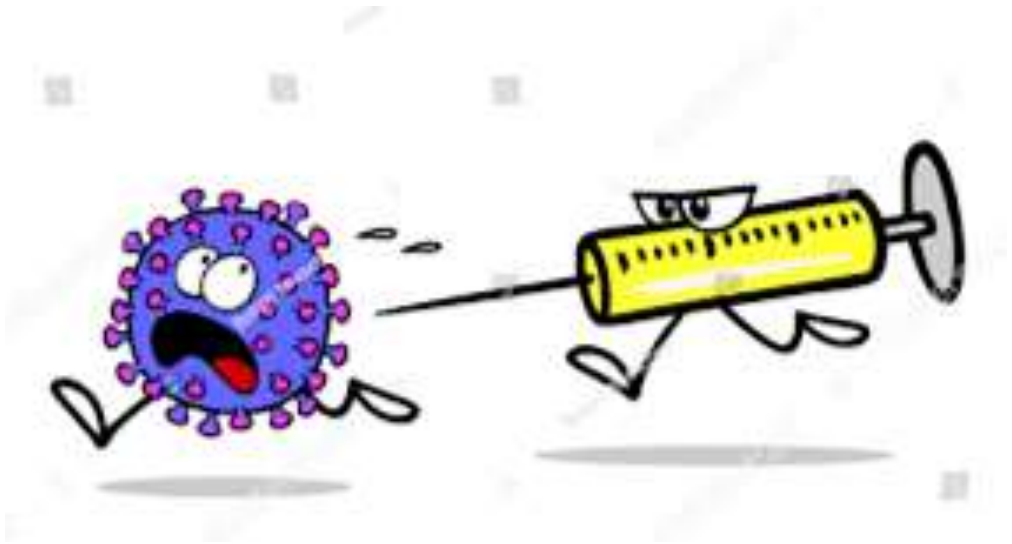
What if inflammation keeps going?

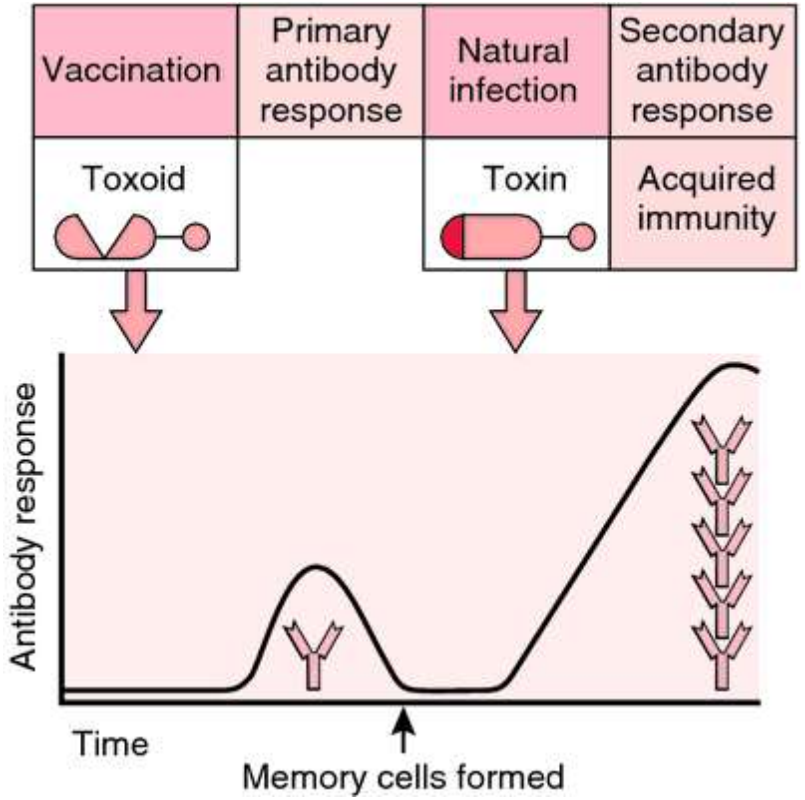




Vaccination

Two key elements: Specificity and memory.





Immunization with tetanus toxoid. Chemical modification of tetanus toxin produces a toxoid which has lost toxicity but retains it’s epitopes.

HOW VACCINES WORK

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- Vaccines contain a component or weakened form of a bacteria or virus. The component or weakened form of bacteria or virus in the vaccine does not cause disease.
- Vaccines teach your body's defence (immune) system how to protect you against a specific type of bacteria or virus.
- Different vaccines protect against different types of bacteria or viruses.

- To ensure you have full and lasting protection, you may need to get more than one dose of some vaccines.

Immune system

TABLE 1. Innate and Adaptive Immunity*

Innate	Adaptive
Nonspecific	Specific
Present at all times	Develops in response to infection
Immediate but general protection	Protection against specific pathogens
Activates adaptive immune response	Leverages components of the innate response
Does not improve with repeated exposure to a pathogen	Memory develops, which may provide lifelong immunity to reinfection with the same pathogen

ANTIGEN VERSUS ANTIBODY	
An antigen is any substance that triggers an immune response in the body	An antibody is a blood protein that is produced against a specific antigen
Also called immunogens	Also called immunoglobulins
Can be either proteins, carbohydrates, lipids or nucleic acids	Glycoproteins
Interacting domain with the antibody is called the epitope	Variable site can bind to the epitope
Cause either diseases or allergic reactions	Protect the body from antigens either by immobilizing the antigen or lysing the pathogen
The four types include exogenous antigens, endogenous antigens, autoantigens, and neoantigens	The four types include IgM, IgG, IgE, IgD, and IgA

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